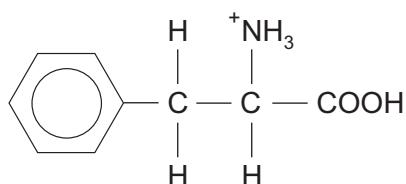


4. (a) In acid solution the formula of the ion formed from phenylalanine is



Give the formula of the ion formed from phenylalanine in basic solution.

[1]

- (b) The melting temperatures of four compounds are shown in the table.

Compound	Melting temperature / °C
CH ₃ CH ₂ COOH	-21
CH ₃ CH(OH)COOH	18
CH ₃ CHBrCOOH	26
CH ₃ CH(NH ₂)COOH	289

Explain why the melting temperature of 2-aminopropanoic acid is very much higher than the other acids in the table.

[2]

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